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Errors

- Programs can contain errors at different levels.
- Programs can be:
    1. **Lexical errors:** such as misspelling of identifier, keyword or operator.
    2. **Syntactic errors:** such as arithmetic expressions with unbalanced parenthesis.
    3. **Semantic errors:** such as an operator applied to an incompatible operand.
    4. **Logical errors:** such as infinitely recursive call due to transfer of control to a block that does not exist.

Top-Down Parsing

- It can be viewed as an attempt to construct a parse tree for the input starting from the root and creating the nodes of the parse tree in pre-order.

Recursive Descent Parsing

- It is a top-down method of syntax analysis in which we execute a set of recursive procedures to process the input.
- Predictive parsing is a special form of recursive descent parsing in which the lookahead symbol unambiguously determines the procedure selected for each non-terminal.

Functions for Parsing

1. **FIRST**
    - If α is any string of grammar symbols, then $FIRST(\alpha)$ is the set of terminals that begins the strings derived from α, i.e. the set of terminals that can appear at the far left of any parse tree derived from a non-terminal.
    - If $\alpha \Rightarrow^* \epsilon$, then $\epsilon \in FIRST(\alpha)$.
    - Rules for finding $FIRST(X)$:
        1. If X is a terminal, then $FIRST(X)$ is $\{X\}$.
        2. If X is a non-terminal and there is a production of the form $X \to Y_1 Y_2 \ldots Y_k$, then place a in $FIRST(X)$ if for some i, a is in $FIRST(Y_i)$, and ϵ is in all of $FIRST(Y_1)$, $FIRST(Y_2)$, $\ldots$ $FIRST(Y_{i-1})$. If ϵ is in all of $FIRST(Y_i)$ from 1 to k, then also add ϵ to $FIRST(X)$.
        3. If $A \to \epsilon$, then add ϵ to $FIRST(A)$.

2. **FOLLOW**

- We define $FOLLOW(A)$ for non-terminal A to be the set of terminals a that can appear immediately to the right of A in some sentential form, i.e. the set of terminals a such that there exists a derivation of the form $S \Rightarrow^* \alpha A a \beta$ for some α and β.
- Rules for finding FOLLOW:
  1. Place $\$$ in $FOLLOW(S)$ where S is the start symbol and $\$$ is the input end marker.
  2. If there is a production, $A \rightarrow \alpha B \beta$, then everything in $FIRST(\beta)$ except for ϵ is placed in $FOLLOW(B)$.
  3. If there is a production $A \rightarrow \alpha B$ or a production $A \rightarrow \alpha B \beta$, where $FIRST(\beta)$ contains ϵ, i.e. $\beta \Rightarrow^* \epsilon$, then everything in $FOLLOW(A)$ is in $FOLLOW(B)$.
- Little modification (makes it easier to implement):
  1. Add $\$$ to $FOLLOW(S)$.
  2. For each production rule of the form $A \rightarrow B_1 B_2 \ldots B_k$
     1. Add everything from $FOLLOW(A)$ to $FOLLOW(B_k)$.
     2. For each $B_i \in V \cup T$ such that $i = 1, 2, \ldots k-1$
        1. Add everything from $FIRST(B_{i+1} B_{i+2} \ldots B_k)$ except ϵ to $FOLLOW(B_i)$.
        2. If $FIRST(B_{i+1} B_{i+2} \ldots B_k)$ contains ϵ, add everything from $FOLLOW(A)$ to $FOLLOW(B_i)$.
  - Note that add everything from $FOLLOW(X)$ to $FOLLOW(Y)$ operations are to deferred till the end because X may not be completely filled at the stage it was called (best to keep a reference to $FOLLOW(X)$ and not the actual values).

Example 1

Consider a grammar with the following production rules:

1. $E \rightarrow TE'$
2. $E' \rightarrow +TE' \mid \epsilon$
3. $T \rightarrow FT'$
4. $T' \rightarrow *FT' \mid \epsilon$
5. $F \rightarrow (E) \mid id$

$FIRST(E) = FIRST(T) = FIRST(F) = \{(,\ id\}$
$FIRST(E') = \{+,\ \epsilon\}$
$FIRST(T') = \{*,\ \epsilon\}$

Finding FOLLOW:

- Add $\$$ to $FOLLOW(E)$.
- From production 1:
  - Add everything from $FOLLOW(E)$ to $FOLLOW(E')$.
  - Add everything from $FIRST(E')$ to $FOLLOW(T)$.

    - Since $FIRST(E')$ contains ϵ, add everything from $FOLLOW(E)$ to $FOLLOW(T)$.
- From production 2:
    - Add everything from $FOLLOW(E')$ to $FOLLOW(T)$.
- From production 3:
    - Add everything from $FOLLOW(T)$ to $FOLLOW(T')$.
    - Add everything from $FIRST(T')$ to $FOLLOW(F)$.
    - Since $FIRST(T')$ contains ϵ, add everything from $FOLLOW(T)$ to $FOLLOW(F)$.
- From production 4:
    - Add everything from $FOLLOW(T')$ to $FOLLOW(F)$.
- From production 5:
    - Add $)$ to $FOLLOW(E)$.

$FOLLOW(E) = FOLLOW(E') = \{\$,\ )\}$
$FOLLOW(T) = FOLLOW(T') = \{+,\ \$,\ )\}$
$FOLLOW(F) = \{*,\ +,\ \$,\ )\}$

Example 2

Consider a grammar with the following production rules:

1. $S \to iEtSS' \mid a$
2. $S' \to eS \mid \epsilon$
3. $E \to b$

$FIRST(S) = \{i,\ a\}$
$FIRST(S') = \{e,\ \epsilon\}$
$FIRST(E) = \{b\}$

Finding FOLLOW:

- Add $\$$ to $FOLLOW(S)$.
- From production 1:
    - Add everything from $FOLLOW(S)$ to $FOLLOW(S')$.
    - Add t to $FOLLOW(E)$.
    - Add everything from $FIRST(S')$ to $FOLLOW(S)$.
- From production 2:
    - Add everything from $FOLLOW(S')$ to $FOLLOW(S)$.

$FOLLOW(S) = \{\$,\ e\}$
$FOLLOW(S') = \{\$,\ e\}$
$FOLLOW(E) = \{t\}$

Example 3

Consider a grammar with the following production rules:

1. $E \rightarrow TE'$
2. $E' \rightarrow +E \mid \epsilon$
3. $T \rightarrow FT'$
4. $T' \rightarrow T \mid \epsilon$
5. $F \rightarrow PF'$
6. $F' \rightarrow *F' \mid \epsilon$
7. $P \rightarrow (E) \mid a \mid b \mid \epsilon$

$FIRST(E) = FIRST(T) = FIRST(F) = FIRST(P) \cup FIRST(F') = \{(,\ a,\ b,\ *,\ \epsilon\}$
$FIRST(E') = \{+,\ \epsilon\}$
$FIRST(T') = FIRST(T) = \{(,\ a,\ b,\ *,\ \epsilon\}$
$FIRST(P) = \{(,\ a,\ b,\ \epsilon\}$
$FIRST(F') = \{*,\ \epsilon\}$

Finding FOLLOW:

- Add $\$$ to $FOLLOW(E)$.
- From production 1:
  - Add everything from $FOLLOW(E)$ to $FOLLOW(E')$.
  - Add everything from $FIRST(E')$ to $FOLLOW(T)$.
  - Since $FIRST(E')$ contains ϵ, add everything from $FOLLOW(E)$ to $FOLLOW(T)$.
- From production 2:
  - Add everything from $FOLLOW(E')$ to $FOLLOW(E)$.
- From production 3:
  - Add everything from $FOLLOW(T)$ to $FOLLOW(T')$.
  - Add everything from $FIRST(T')$ to $FOLLOW(F)$.
  - Since $FIRST(T')$ contains ϵ, add everything from $FOLLOW(T)$ to $FOLLOW(F)$.
- From production 4:
  - Add everything from $FOLLOW(T')$ to $FOLLOW(T)$.
- From production 5:
  - Add everything from $FOLLOW(F)$ to $FOLLOW(F')$.
  - Add everything from $FIRST(F')$ to $FOLLOW(P)$.
  - Since $FIRST(F')$ contains ϵ, add $FOLLOW(F)$ to $FOLLOW(P)$.
- From production 7:
  - Add) to $FOLLOW(E)$.

$FOLLOW(E) = FOLLOW(E') = \{\$,\ )\}$
$FOLLOW(T) = FOLLOW(T') = \{+,\ \$,\ )\}$
$FOLLOW(F) = FOLLOW(F') = FOLLOW(P) = \{(,\ a,\ b,\ *,\ +,\ \$,\ )\}$